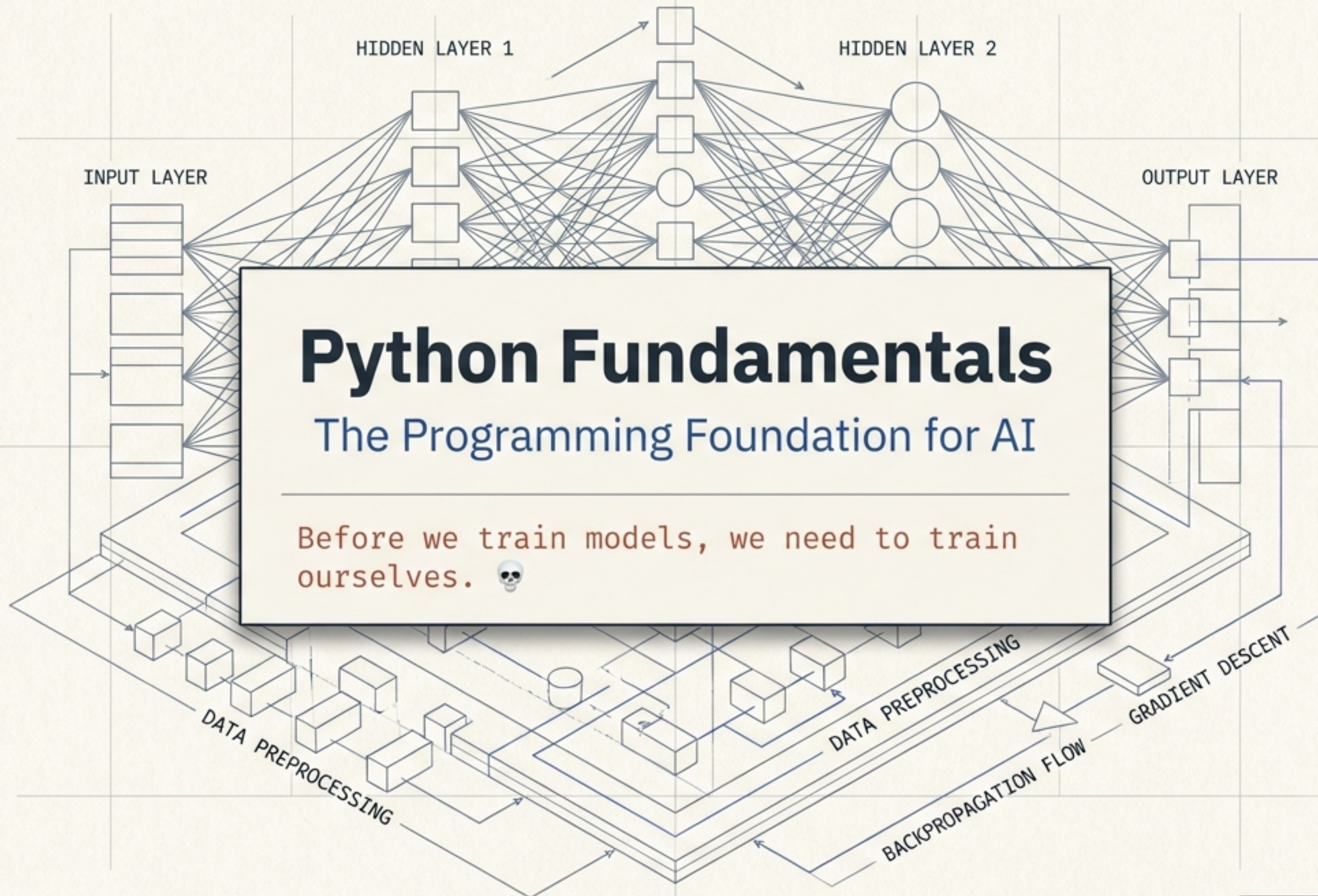




Giving the System a Voice

```
print("Hello, AI")
```

The execution command:
Used to display internal
machine state.

Hello, AI

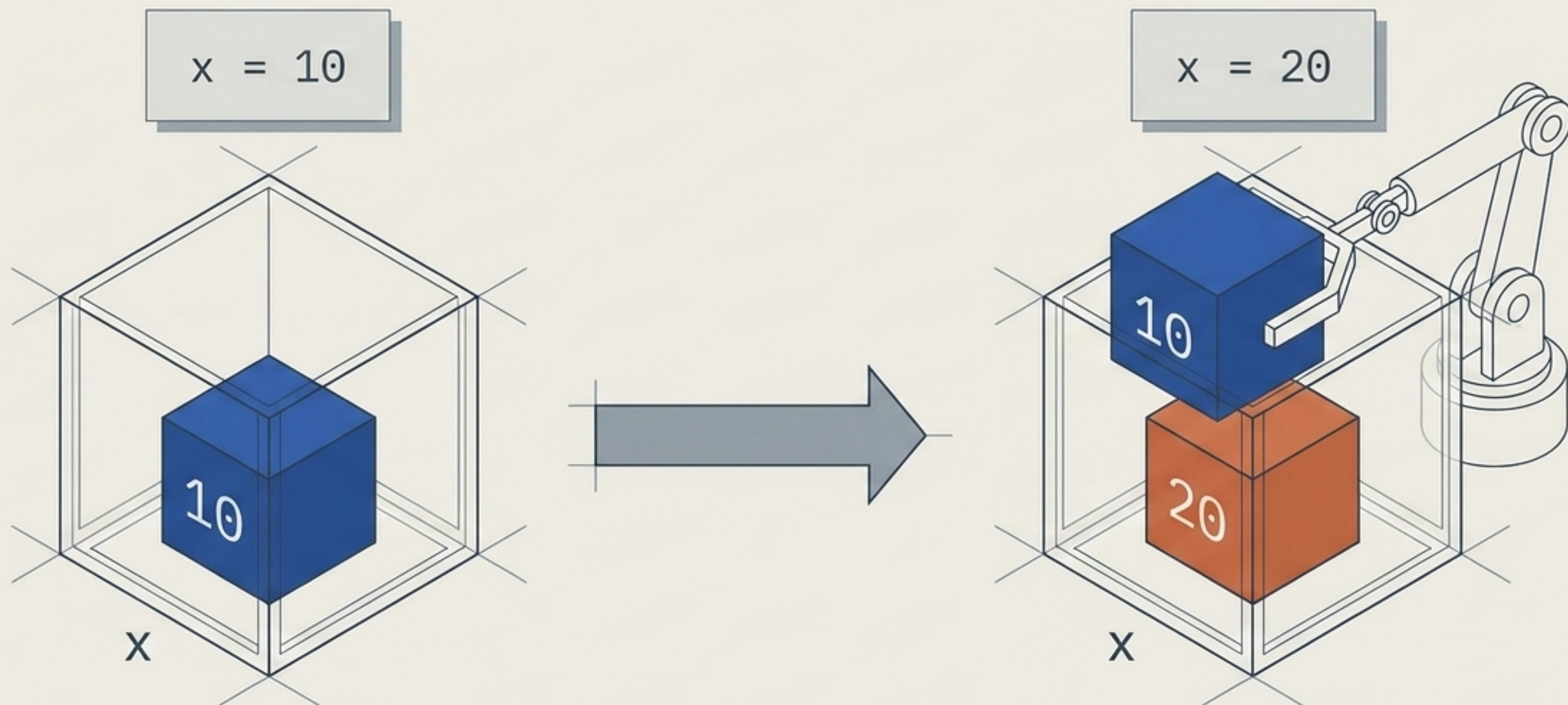
Machine
calculates value

print() invoked

Value displayed
to engineer

Allocating the Machine's Short-Term Memory

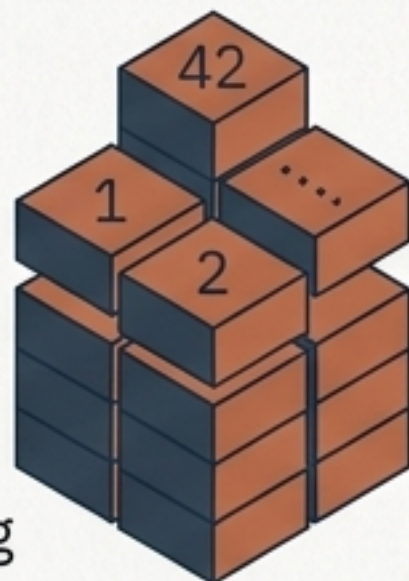
A variable is simply a name that refers to a dynamically changing value.



The Four Core Materials of AI Data

Integer

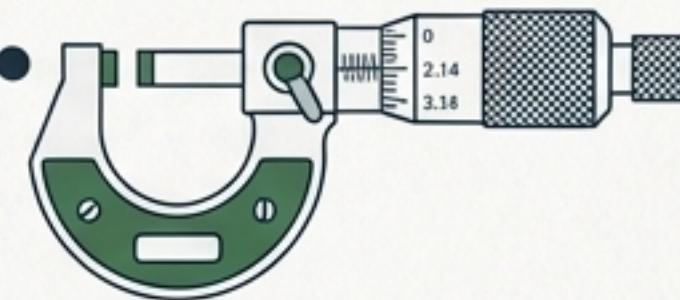
```
...  
epochs = 42
```



AI Application: Counting Training Epochs or Iterations.

Float

```
...  
weight = 3.14
```



AI Application: Model Weights, Probabilities, and Confidence Scores.

String

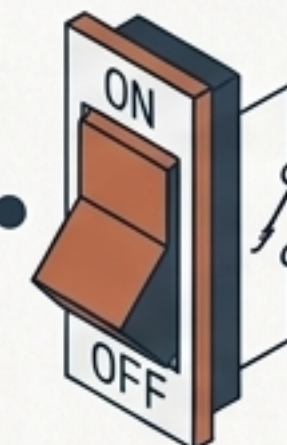
```
...  
prompt = "Hello"
```



AI Application: Natural Language Processing (NLP) prompts and text data.

Boolean

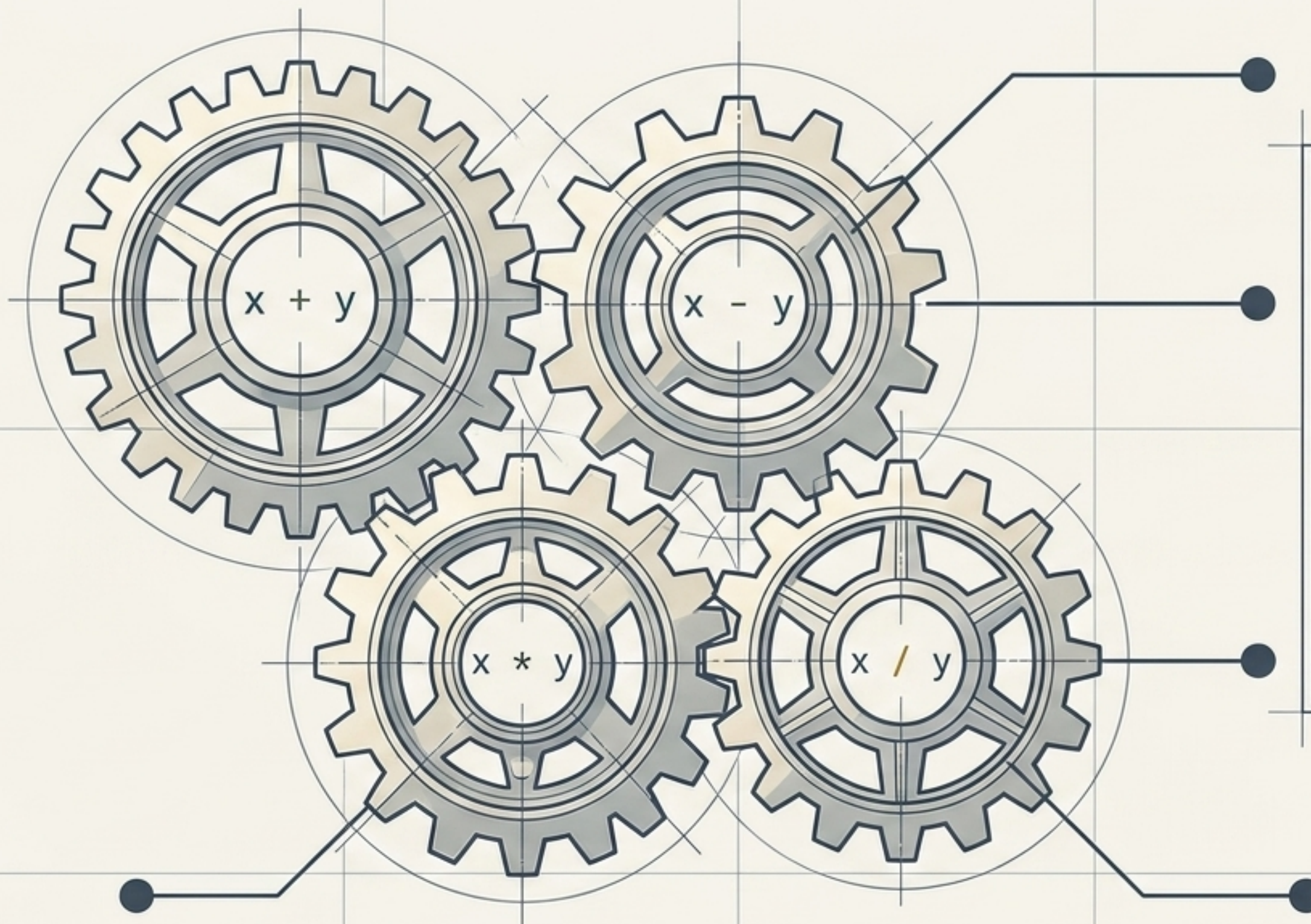
```
...  
is_active = True
```



AI Application: Activation States and binary flags.

AI works heavily with numerical data. Libraries like NumPy will let us manipulate huge collections of these exact mathematical values.

The Engine of Mathematical Models



Python supports standard mathematical operations natively.

These basic calculations act as the foundational gears driving all complex machine learning models, from simple regressions to deep neural networks.

Evaluating State and Machine Logic

Comparing values always resolves to a Boolean.

10 > 5

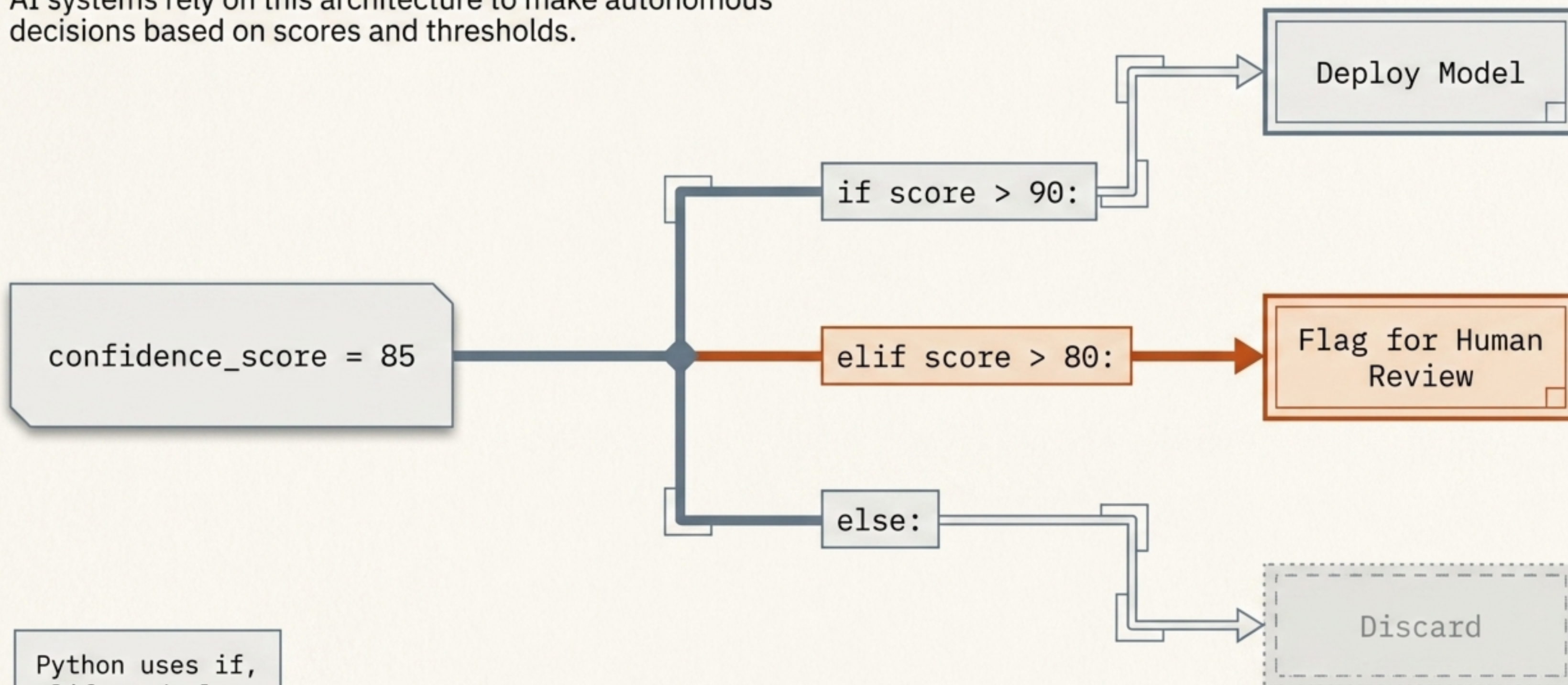


True

==	!=	>	<	>=	<=
==	≠	>	<	≥	≤
Equal	Not Equal	Greater	Less	Greater/Equal	Less/Equal

Routing Data with Conditional Decisions

AI systems rely on this architecture to make autonomous decisions based on scores and thresholds.

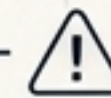


Python uses `if`,
`elif`, and `else`

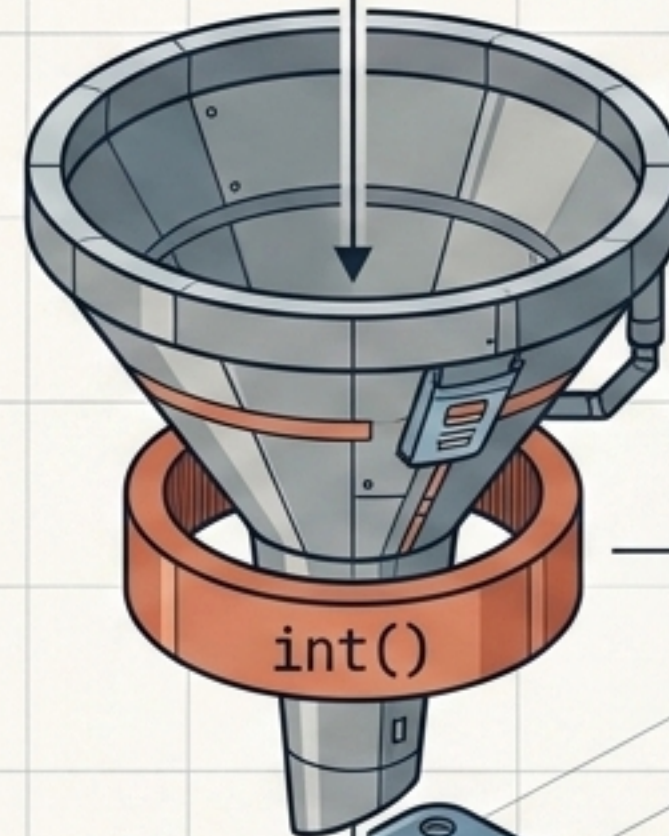
Ingesting and Transforming Human Input

```
age = input("Enter age: ") 25|
```

"25"



input() defaults to String!



Data must be explicitly cast into numbers.

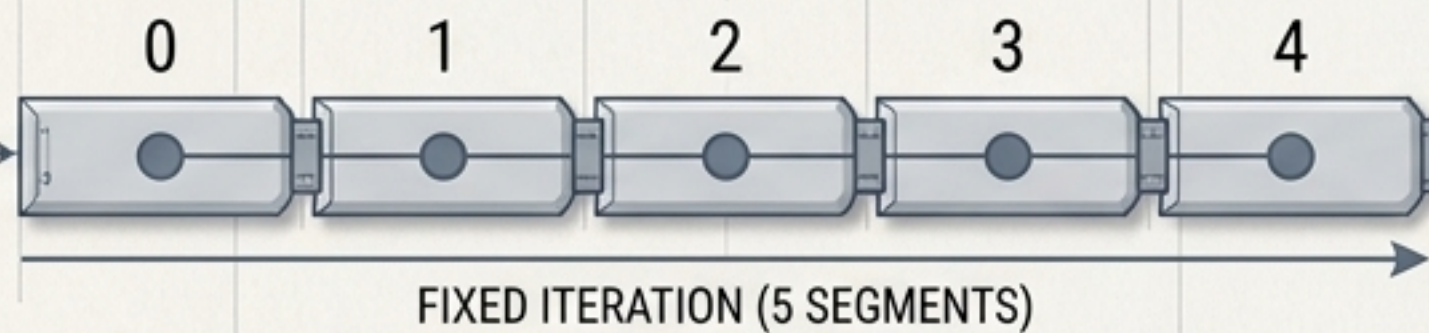
25

Ready for AI Computation

Automating Repetition at Scale

for

```
for i in range(5):  
    print(i)
```



Used for known sequences, like running a set number of training epochs over a dataset.

while

```
while accuracy < 0.95:  
    train_model()
```



Used for continuous processing until a specific condition is met or broken.

Organizing Sequential Datasets

```
data = [10, 20, 30]
```

[0]

10

[1]

20

[2]

30

Python uses **Zero-Based Indexing**.
Lists are the foundational precursor to working with massive, one-dimensional arrays in NumPy.

Mapping Structured Knowledge

Dictionaries store data using specific named keys rather than sequential numbers.

```
model = {"name": "GPT", "layers": 12}
```

Key: "name"

Value: "GPT"

The Standard for Structured AI Data



Configuration



JSON



API Responses

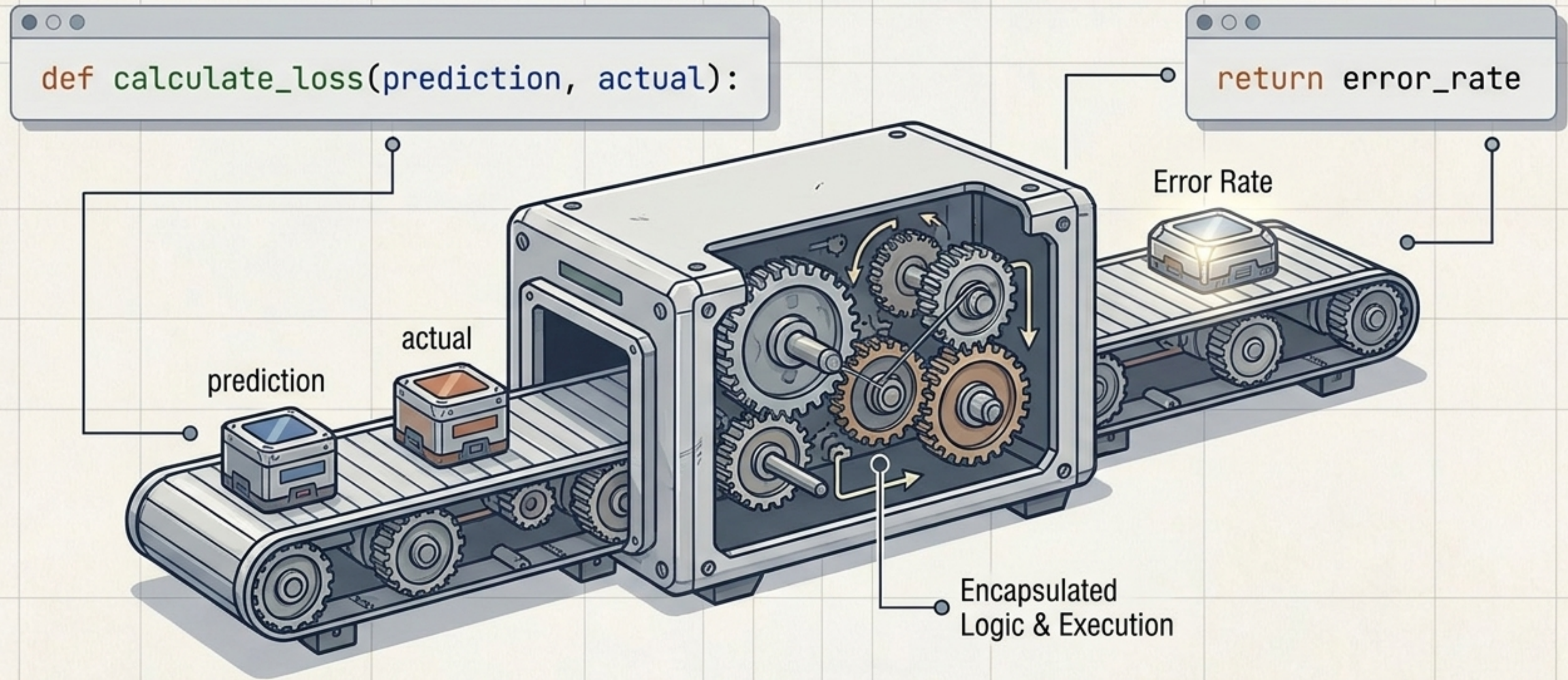


Metadata



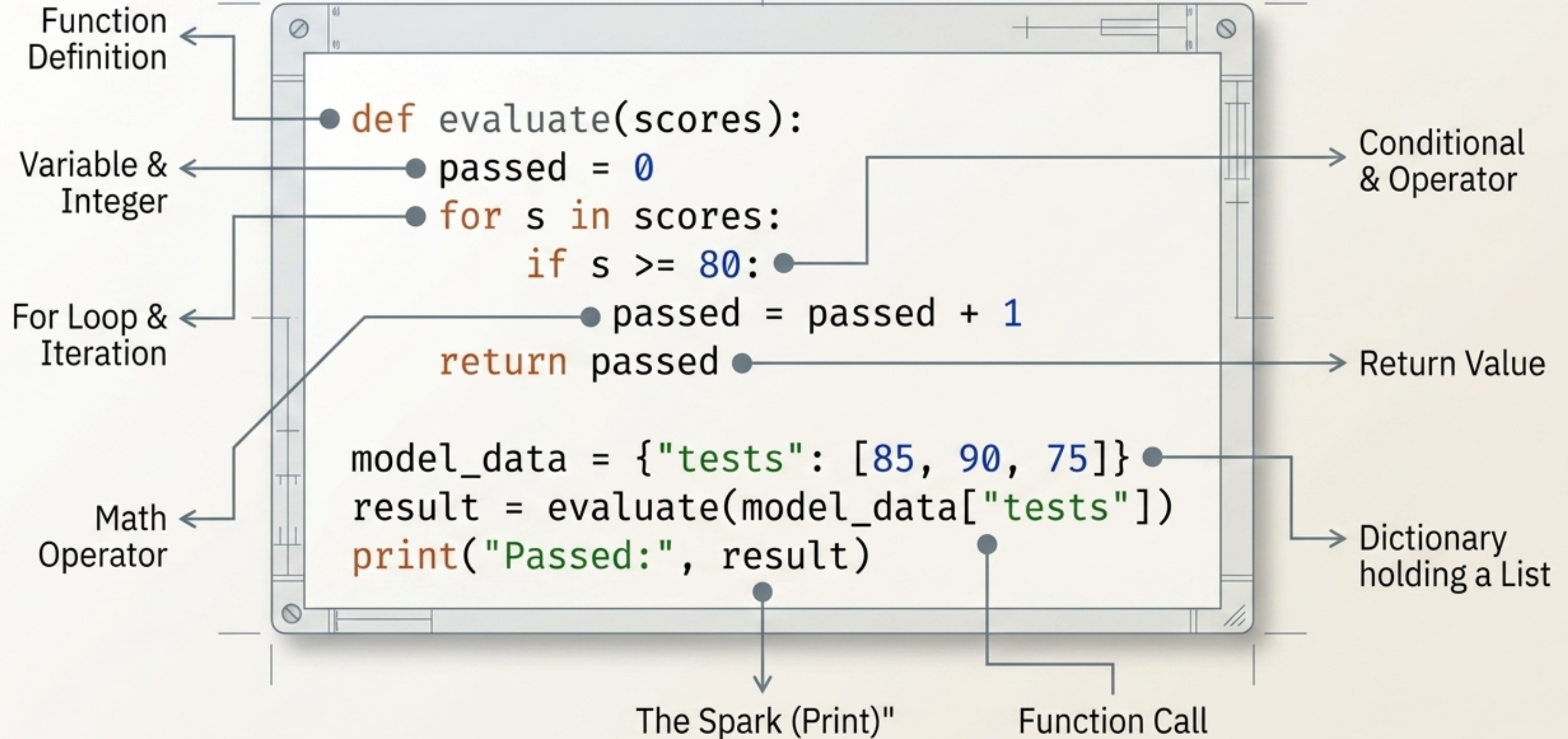
Model Settings

Building Reusable Data Pipelines



Functions are the essential building blocks for constructing modular, automated Machine Learning pipelines.

Anatomy of an AI Evaluation Script



The Rite of Passage



Python Beginner

Why are there four variables?

AI Engineer

You'll understand soon. 🦴



The Foundation is Laid

You now understand the basic building blocks.

These aren't just Python concepts—they are the required foundation for manipulating data and building AI systems with NumPy and Pandas.

Challenge

- `name = input("Name: ")`
- `age = int(input("Age: "))`

Write the code to produce this exact Goal Output:

```
print("AI User", name, "is",  
      age * 2, "in AI years.")
```